Overview of Propositional Logic

- Propositions (T,F) and logical operators ($\neg,\wedge,\vee,\oplus,\rightarrow,\leftrightarrow$)
- A **tautology** is a compound proposition that is always true (e.g. $p \vee \neg p$).
- A **contradiction** is a compound proposition that is always false (e.g. $p \wedge \neg p$).
- The compound propositions p and q are called **logically equivalent** if $p \leftrightarrow q$ is a tautology.
  - The notation is $p \equiv q$ (or $p \Leftrightarrow q$).

Propositional Equivalences

Equivalence	Name
$p \wedge \mathsf{T} \equiv p$ $p \vee \mathsf{F} \equiv p$	Identity laws
$p \vee \mathsf{T} \equiv \mathsf{T}$ $p \wedge \mathsf{F} \equiv \mathsf{F}$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws

Propositional Equivalences (cont'd)

Equivalence	Name
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathsf{T}$ $p \wedge \neg p \equiv \mathsf{F}$	Negation laws

Propositional Equivalences (cont'd)

Equivalence	Name
$p \to q \equiv \neg p \lor q$ $p \to q \equiv \neg q \to \neg p$ $(p \to q) \land (p \to r) \equiv p \to (q \land r)$ $(p \to q) \lor (p \to r) \equiv p \to (q \lor r)$ $(p \to r) \land (q \to r) \equiv (p \lor q) \to r$ $(p \to r) \lor (q \to r) \equiv (p \land q) \to r$	Involving conditional statements
$p \leftrightarrow q \equiv (p \to q) \land (q \to p)$ $p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$ $p \leftrightarrow q \equiv (p \land q) \lor (\neg p \land \neg q)$	Involving biconditional statements

Logic Puzzle

- Determine the relative salaries of three coworkers using two facts.

1. If Fred is not the highest paid of the three, then Janice is.
2. If Janice is not the lowest paid, then Maggie is paid the most.

- Determine the relative salaries of Fred, Maggie, and Janice.

Notation: (F=1) denotes the proposition that Fred is the highest paid, (J=3) Janice is the lowest paid of the three etc.

1. I interpret the first fact as $(F=1) \oplus (J=1)$ because either (F=1) or (J=1) is true, but not both since only one person can be the most highly paid.
2. I interpret the second fact as $\neg(J=3) \rightarrow (M=1) \equiv (J=3) \vee (M=1)$.

Assume that (J=1). Then the second fact cannot be true because $M \neq 1$ and $J \neq 3$.

Therefore, it must be that (F=1). For the second fact to be truth, it must hold that (J=3).

Thus, the (decreasing) order of salaries is Fred, Maggie and Janice.

Differences between human and propositional logic

A real experiment was performed many years ago to study the relationship between human and propositional logic.

Human subjects were given the rule: "If the envelope is sealed, it must contain a stamp on the other side", and were asked to turn the minimum number of envelopes in order to verify the rule.

An alternative description: you work in the post-office and you want to make sure that **all sealed envelopes have stamps**, by turning the minimum number of envelopes. There are 4 cases:

1: You are shown a sealed envelope – do you need to turn it to check whether it contains a stamp?

Yes. I turn it over to make sure that it contains a stamp.

In terms of propositional logic you can think of

a] p as the proposition "the envelope is sealed",

b] q as the proposition "the envelope contains a stamp".

c] The rule is $p \rightarrow q$

Since p is true, you must check that q is also true.

Differences between human and propositional logic (cont)

2: You are shown an open envelope – do you need to turn it to check whether it contains a stamp?

No. The rule does not apply to open envelopes. They may contain a stamp or not. Since I have $\neg p$, I do not care about q.

3: You are shown an envelope with a stamp – do you need to turn it to check whether it is sealed?

No. I do not care whether the envelope is sealed or open. Since I have q, I do not care about whether it is the case of p or $\neg p$.

4: You are shown an envelope without a stamp – do you need to turn it to check whether it is sealed?

Yes. The envelope must be open, otherwise there is a rule violation. Since I have $\neg q$, it must be $\neg p$ (**contrapositive:** $\neg q \rightarrow \neg p$).

The majority of human subjects gave the wrong answer for case 4.

Overview of Predicate Logic

- Predicates and Quantifiers (∀,∃)
- Predicates have variables that take values from a domain
- All variables must be bound by a quantifier (or set to a particular value).
- When the domain has *n* elements $x_1, x_2, \ldots, x_n$

  $\forall x\, P(x)$ is the same as $P(x_1) \wedge P(x_2) \wedge \ldots \wedge P(x_n)$,

  $\exists x\, P(x)$ is the same as $P(x_1) \vee P(x_2) \vee \ldots \vee P(x_n)$
- The quantifiers ∀ and ∃ have higher precedence than all logical operators.
- When the scopes of two variables are disjoint we can use the same (or different) names
  - $\exists x P(x) \vee \forall x Q(x)$ is the same as $\exists x P(x) \vee \forall y Q(y)$
    - Better to use different variable names to avoid confusion

Quantifier with Restricted Domain

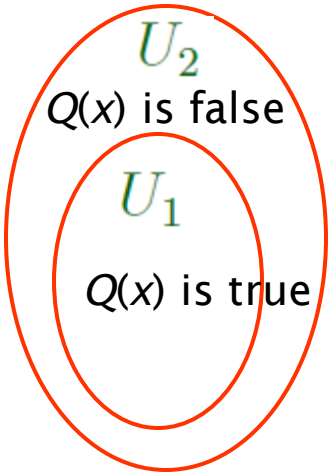
- Let U_1 and U_2 be two domains with $U_1 \subseteq U_2$. Suppose U_1 is defined as:

$$U_1 = \{\ x \in U_2 \mid Q(x) \text{ is true } \}.$$

- Then, a statement $P(x)$ about U_1 may also be interpreted as a statement about U_2

(a) $\forall x \in U_1\ (P(x))$ is equivalent to
$\forall x \in U_2\ (Q(x) \rightarrow P(x))$

(b) $\exists x \in U_1\ (P(x))$ is equivalent to
$\exists x \in U_2\ (Q(x) \wedge P(x))$



Exercise on Restricted Domains

- Let $Q(x)$:"x is a lion", $P(x)$:"x is fierce", and $R(x)$:"x drinks coffee", respectively.

  Assume that the domain U_2 contains all creatures, and U_1 contains only lions

- Consider the following argument:

1. Premise 1: "All lions are fierce"
2. Premise 2: "Some lions do not drink coffee"
3. Conclusion: "Some fierce creatures do not drink coffee"

- Express the statements using quantifiers and $P(x)$, $Q(x)$, and $R(x)$.

1. $\forall x\,(Q(x) \rightarrow P(x))$
2. $\exists x\,(Q(x) \wedge \neg R(x))$
3. $\exists x\,(P(x) \wedge \neg R(x))$

- **Remark:** We will later discuss the issue of determining whether the conclusion is valid. In this example, it is.

De Morgan's Laws for Quantifiers

De Morgan's Laws for Quantifiers			
Negation	Equivalent statement	When is it true?	When is it false?
$\neg\forall x\, P(x)$	$\exists x\, \neg P(x)$	There exists an x for which $P(x)$ is false.	$P(x)$ is true for every x.
$\neg\exists x\, Q(x)$	$\forall x\, \neg Q(x)$	$Q(x)$ is false for every x.	There exists an x for which $Q(x)$ is true.

Quantifications of Two Variables and Negations

- Rule: Move the negation one level in and switch the quantifier

Quantifications of Two Variables		
Statement	When is it true?	When is it false?
$\forall x \forall y\, P(x,y)$ $\forall y \forall x\, P(x,y)$	$P(x,y)$ is true for every pair x, y.	$\neg\forall x\, \forall y\, (P(x,y)) \equiv \exists x\, \neg\forall y\, (P(x,y)) \equiv \exists x\, \exists y\, (\neg P(x,y))$
$\forall x \exists y\, P(x,y)$	For every x there is a y for which $P(x,y)$ is true.	
$\exists x \forall y\, P(x,y)$	There is an x such that $P(x,y)$ is true for every y.	
$\exists x \exists y\, P(x,y)$ $\exists y \exists x\, P(x,y)$	There is a pair x, y for which $P(x,y)$ is true.	

Quantifications of Two Variables

- Rule: Move the negation one level in and switch the quantifier

Quantifications of Two Variables		
Statement	When is it true?	When is it false?
$\forall x \forall y\, P(x,y)$ $\forall y \forall x\, P(x,y)$	$P(x,y)$ is true for every pair x, y.	There is a pair x, y for which $P(x,y)$ is false.
$\forall x \exists y\, P(x,y)$	For every x there is a y for which $P(x,y)$ is true.	$\neg\forall x\, \exists y\, (P(x,y)) \equiv \exists x\, \neg\exists y\, (P(x,y)) \equiv \exists x\, \forall y\, (\neg P(x,y))$
$\exists x \forall y\, P(x,y)$	There is an x such that $P(x,y)$ is true for every y.	
$\exists x \exists y\, P(x,y)$ $\exists y \exists x\, P(x,y)$	There is a pair x, y for which $P(x,y)$ is true.	

Quantifications of Two Variables

- Rule: Move the negation one level in and switch the quantifier

Quantifications of Two Variables		
Statement	When is it true?	When is it false?
$\forall x \forall y\, P(x,y)$ $\forall y \forall x\, P(x,y)$	$P(x,y)$ is true for every pair x, y.	There is a pair x, y for which $P(x,y)$ is false.
$\forall x \exists y\, P(x,y)$	For every x there is a y for which $P(x,y)$ is true.	There is an x such that $P(x,y)$ is false for every y.
$\exists x \forall y\, P(x,y)$	There is an x such that $P(x,y)$ is true for every y.	$\neg\exists x\, \forall y\, (P(x,y)) \equiv \forall x\, \neg\forall y\, (P(x,y)) \equiv \forall x\, \exists y\, (\neg P(x,y))$
$\exists x \exists y\, P(x,y)$ $\exists y \exists x\, P(x,y)$	There is a pair x, y for which $P(x,y)$ is true.	

Quantifications of Two Variables

- The following table summarizes the meanings of the different possible quantifications involving two variables.

Quantifications of Two Variables		
Statement	When is it true?	When is it false?
$\forall x \forall y\, P(x,y)$ $\forall y \forall x\, P(x,y)$	$P(x,y)$ is true for every pair x, y.	There is a pair x, y for which $P(x,y)$ is false.
$\forall x \exists y\, P(x,y)$	For every x there is a y for which $P(x,y)$ is true.	There is an x such that $P(x,y)$ is false for every y.
$\exists x \forall y\, P(x,y)$	There is an x such that $P(x,y)$ is true for every y.	For every x there is a y for which $P(x,y)$ is false.
$\exists x \exists y\, P(x,y)$ $\exists y \exists x\, P(x,y)$	There is a pair x, y for which $P(x,y)$ is true.	

$\neg\exists x\, \exists y\, (P(x,y)) \equiv \forall x\, \neg\exists y\, (P(x,y)) \equiv \forall x\, \forall y\, (\neg P(x,y))$

Quantifications of Two Variables

- Rule: Move the negation one level in and switch the quantifier

Quantifications of Two Variables		
Statement	When is it true?	When is it false?
$\forall x \forall y\, P(x,y)$ $\forall y \forall x\, P(x,y)$	$P(x,y)$ is true for every pair x, y.	There is a pair x, y for which $P(x,y)$ is false.
$\forall x \exists y\, P(x,y)$	For every x there is a y for which $P(x,y)$ is true.	There is an x such that $P(x,y)$ is false for every y.
$\exists x \forall y\, P(x,y)$	There is an x such that $P(x,y)$ is true for every y.	For every x there is a y for which $P(x,y)$ is false.
$\exists x \exists y\, P(x,y)$ $\exists y \exists x\, P(x,y)$	There is a pair x, y for which $P(x,y)$ is true.	$P(x,y)$ is false for every pair x, y.

Logical Equivalences

- We can distribute a universal quantifier over a conjunction.

$$\forall x\,(Q(x) \wedge P(x)) \;\equiv\; \forall x\,P(x) \wedge \forall x\,Q(x)$$

- We can also distribute an existential quantifier over a disjunction:

$$\exists x\,(P(x) \vee Q(x)) \;\equiv\; \exists x\,P(x) \vee \exists x\,Q(x)$$

However, we cannot distribute a universal quantifier over a disjunction, nor can we distribute an existential quantifier over a conjunction:

$$\forall x\,(P(x) \vee Q(x)) \;\not\equiv\; \forall x\,P(x) \vee \forall x\,Q(x)$$

$$\exists x\,(P(x) \wedge Q(x)) \;\not\equiv\; \exists x\,P(x) \wedge \exists x\,Q(x)$$

Counterexample 1: we cannot distribute a universal quantifier over a disjunction

- Show with a counter example that $\forall x\,(P(x) \vee Q(x))$ is not equivalent to $\forall x\, P(x) \vee \forall x\, Q(x)$.
- Note: recall that $\forall x\, P(x) \vee \forall x\, Q(x) \equiv \forall x\, P(x) \vee \forall y\, Q(y)$
  - When the scopes of two variables are disjoint we can use different variables
- Let $P(x)$ the predicate "x is male"
- Let $Q(x)$ the predicate "x is female"
- Let the domain of x be all human.
- $\forall x\,(P(x) \vee Q(x))$ means that each human is male or female (true).
- $\forall x\, P(x) \vee \forall y\, Q(y)$ means all humans are male or all humans are female (false)

Counterexample 2: we cannot distribute a existential quantifier over a conjunction

Assume that the domain of x is the set of integers and the predicates $P(x)$: $2x + 1 = 5$
$Q(x)$: $x^2 = 9$

- $\exists x\, P(x) \wedge \exists x\, Q(x)$ (equivalent: $\exists x\, P(x) \wedge \exists y\, Q(y)$)
  - is true because $P(2)$ and $Q(3)$ are true.
- $\exists x\, (P(x) \wedge Q(x))$
  - is false because there is no one integer a such that $P(a)$ and $Q(a)$ are both true.
- Therefore: $\exists x\, P(x) \wedge \exists x\, Q(x)$ is NOT equivalent to $\exists x\, (P(x) \wedge Q(x))$

Null Quantifications: How can we change the scope of quantifiers?

Rules: If variable x does not appear in A (A is a logical expression bound by a quantifier whose scope does not cover x, e.g., $\forall y\ Q(y)$)

Rule 1.

$$\forall x\ (A \wedge P(x)) \equiv A \wedge \forall x\ P(x) \equiv \forall x\ P(x) \wedge A \equiv \forall x\ (P(x) \wedge A)$$

Rule 2.

$$\forall x\ (A \vee P(x)) \equiv A \vee \forall x\ P(x) \equiv \forall x\ P(x) \vee A \equiv \forall x\ (P(x) \vee A)$$

Rule 3.

$$\exists x\ (A \wedge P(x)) \equiv A \wedge \exists x\ P(x) \equiv \exists x\ P(x) \wedge A \equiv \exists x\ (P(x) \wedge A)$$

Rule 4.

$$\exists x\ (A \vee P(x)) \equiv A \vee \exists x\ P(x) \equiv \exists x\ P(x) \vee A \equiv \exists x\ (P(x) \vee A)$$

Proof or rules

- Proof: Let the domain of x be $\{x_1, \ldots, x_n\}$
- Rule 1: $\forall x\ (A \wedge P(x)) \equiv A \wedge \forall x\ P(x)$

$\forall x\ (A \wedge P(x)) \equiv (A \wedge P(x_1)) \wedge \cdots \wedge (A \wedge P(x_n)) \equiv$
$A \wedge P(x_1) \wedge \cdots \wedge P(x_n) \equiv A \wedge \forall x\ P(x)$

- Rule 2: $\forall x\ (A \vee P(x)) \equiv A \vee \forall x\ P(x)$

$\forall x\ (A \vee P(x)) \equiv (A \vee P(x_1)) \wedge \cdots \wedge (A \vee P(x_n)) \equiv$
$A \vee (P(x_1) \wedge \cdots \wedge P(x_n)) \equiv A \vee \forall x\ P(x)$

- Rule 3: $\exists x\ (A \wedge P(x)) \equiv A \wedge \exists x\ P(x)$

$\exists x\ (A \wedge P(x)) \equiv (A \wedge P(x_1)) \vee \cdots \vee (A \wedge P(x_n)) \equiv$
$A \wedge (P(x_1) \vee \cdots \vee P(x_n)) \equiv A \wedge \exists x\ P(x)$

- Rule 4: $\exists x\ (A \vee P(x)) \equiv A \vee \exists x\ P(x)$

$\exists x\ (A \vee P(x)) \equiv (A \vee P(x_1)) \vee \cdots \vee (A \vee P(x_n)) \equiv$
$(A \vee ((P(x_1) \vee \cdots \vee P(x_n)) \equiv A \vee \exists x\ P(x)$

- Note that all rules hold also for $\neg P(x)$
  - E.g., Rule 1: $\forall x\ (A \wedge \neg P(x)) \equiv A \wedge \forall x\ \neg P(x)$

PNF: Prenex Normal Form

- PNF means that all the quantifiers are in the beginning
- Every formula in predicate calculus can be written in PNF
- When I convert to PNF, I can use Rules 1-4, provided that all quantifiers are not negated, and all operators are ∨ or ∧.

Guidelines

- Use different variable names
  - ∃x P(x) → ∀**x** Q(**x**) should change to ∃x P(x) → ∀**y** Q(**y**)
- If there are operators other than ∨ or ∧, replace them with equivalent expressions that contain only ∨ or ∧
  - ∃x P(x) → ∀y Q(y) should change to ¬∃x P(x) ∨ ∀**y** Q(**y**)
- Make sure that none of the quantifiers is negated
  - ¬∃x P(x) ∨ ∀**y** Q(**y**) should change to ∀x ¬P(x) ∨ ∀**y** Q(**y**)
- Now you are ready to apply Rules 1-4
  - If you extend the scope of ∀x, and then ∀y you get ∀x∀y (¬P(x) ∨ Q(y))
  - If you extend the scope of ∀y, and then ∀x you get ∀y∀x (¬P(x) ∨ Q(y))
- ∃x P(x) → ∀**y** Q(**y**) ≡ ∀x∀y (¬P(x) ∨ Q(y)) ≡ ∀x∀y (P(x) → Q(y))

PNF Exercise

Convert $\exists z(\exists y\ R(y,z) \rightarrow \forall y\ Q(y,z))$ to PNF.

$\exists z(\exists y\ R(y,z) \rightarrow \forall y\ Q(y,z)) \equiv$ variable change from y to x

$\exists z(\exists y\ R(y,z) \rightarrow \forall x\ Q(x,z)) \equiv$ implication equivalence

$\exists z(\neg\exists y\ R(y,z) \vee \forall x\ Q(x,z)) \equiv$ De Morgan

$\exists z(\forall y\ \neg R(y,z) \vee \forall x\ Q(x,z)) \equiv$ Rule 2 on scope of $\forall x$ (A is $\forall y\ \neg R(y,z)$)

$\exists z\forall x(\forall y\ \neg R(y,z) \vee Q(x,z)) \equiv$ Rule 2 on scope of $\forall y$ (A is $Q(x,z)$)

$\exists z\forall x\forall y(\neg R(y,z) \vee Q(x,z)) \equiv$ already PNF, implication equivalence

$\exists z\forall x\forall y(R(y,z) \rightarrow Q(x,z))$

- I can always reorder quantifiers of the same type
- $\exists z\forall x\forall y(R(y,z) \rightarrow Q(x,z)) \equiv \exists z\forall y\forall x(R(y,z) \rightarrow Q(x,z))$
  - In this example it depends on which scope, $\forall x$ or $\forall y$, I extend first

- I cannot always reorder quantifiers of different type.

PNF: A Counter-intuitive Equivalence

- Recall that $\forall x\,(P(x) \vee Q(x)) \not\equiv \forall x\,P(x) \vee \forall x\,Q(x)$
  - We cannot distribute a universal quantifier over a disjunction
- Lets generate the PNF of the RHS

$\forall x\,P(x) \vee \forall x\,Q(x) \equiv$ change variable of Q

$\forall x\,P(x) \vee \forall y\,Q(y) \equiv$ apply rule 2 to change scope of x

$\forall x\,(P(x) \vee \forall y\,Q(y)) \equiv$ apply rule 2 to change scope of y

$\forall x \forall y\,(P(x) \vee Q(y))$

- Thus, $\forall x\,P(x) \vee \forall x\,Q(x) \equiv \forall x \forall y\,(P(x) \vee Q(y))$
- $\forall x \forall y\,(P(x) \vee Q(y))$ NOT EQUIVALENT to $\forall x\,(P(x) \vee Q(x))$
- The domain contains only two elements a and b such that $P(a)$ is T, $Q(a)$ is F, $P(b)$ is F, $Q(b)$ is T.
- $\forall x\,(P(x) \vee Q(x))$ is T because both $P(a) \vee Q(a)$ and $P(b) \vee Q(b)$ are T.
- $\forall x \forall y\,(P(x) \vee Q(y))$ is False because the pair $P(b) \vee Q(a)$ is F.

Equivalent or Not

- Is $\exists x\ P(x) \rightarrow \exists y\ Q(y)$ EQUIVALENT to $\exists x \exists y\ (P(x) \rightarrow\ Q(y))$?

Consider the domains:

Domain	a	b
P	T	F
Q	F	F

- $\exists x P(x) \rightarrow \exists y\ Q(y) \equiv \neg \exists x P(x) \vee \exists y\ Q(y) \equiv \forall x \neg P(x) \vee \exists y\ Q(y)$ is
- false because both $\forall x\ \neg P(x)$ and $\exists y\ Q(y)$ are F.

- $\exists x \exists y\ (P(x) \rightarrow\ Q(y)) \equiv \exists x \exists y\ (\neg\ P(x) \vee\ Q(y)) \equiv$ **T**

Using the above domains we have True for pairs $(P(b), Q(a))$, $(P(b), Q(b))$ because $\neg\ P(b)$ is T.

- Explanation of non equivalence using PNF of $\exists x\ P(x) \rightarrow \exists y\ Q(y)$:

$\exists x\ P(x) \rightarrow \exists y\ Q(y) \equiv \neg\ \exists x\ P(x) \vee \exists y\ Q(y) \equiv$

$\forall x\ \neg P(x) \vee \exists y\ Q(y) \equiv$ by rule 2 $\forall x\ (\neg P(x) \vee \exists y\ Q(y)) \equiv$ by rule 4

$\forall x \exists y\ (\neg P(x) \vee\ Q(y)) \equiv \mathbf{\forall x \exists y\ (P(x) \rightarrow\ Q(y))}$

Not equivalent to $\exists x \exists y\ (P(x) \rightarrow\ Q(y))$

Equivalent or Not (cont)

- Is $\forall x\ P(x) \rightarrow \exists y\ Q(y)$ EQUIVALENT to $\forall x \exists y\ (P(x) \rightarrow\ Q(y))$?
- $\forall x\ P(x) \rightarrow \exists y\ Q(y) \equiv \neg\ \forall x\ P(x) \vee \exists y\ Q(y) \equiv$ DeMorgan

  $\exists x\ \neg P(x) \vee \exists y\ Q(y)\ \equiv$(applying Rule 4 to the scope of x)

  $\exists x\ (\neg P(x) \vee \exists y\ Q(y)) \equiv$(applying Rule 4 to the scope of y)

  $\exists x \exists y\ (\neg P(x) \vee\ Q(y)) \equiv \exists x \exists y\ (P(x) \rightarrow\ Q(y))$
- $\forall x \exists y\ (P(x) \rightarrow\ Q(y)) \equiv$ DeMorgan

  $\forall x \exists y\ (\neg P(x) \vee\ Q(y)) \equiv$ (applying Rule 4 to the scope of y)

  $\forall x\ \neg P(x) \vee \exists y Q(y) \equiv$ DeMorgan

  $\neg\ \exists x\ P(x) \vee \exists y Q(y) \equiv \exists x\ P(x) \rightarrow \exists y\ Q(y)$
- In the previous slide we established that $\exists x\ P(x) \rightarrow \exists y\ Q(y)$ is NOT EQUIVALENT to $\exists x \exists y\ (P(x) \rightarrow\ Q(y))$

Exercise on Predicate Logic

- Let P(x), and Q(x) be the statements “x is a professor”, “x is ignorant”.
- "All professors are not ignorant" can be expressed as $\forall x\ (P(x) \rightarrow \neg Q(x))$.
- Which of the following statements is equivalent:
    1. $\forall x\ (\neg P(x) \vee \neg Q(x))$
    2. $\neg \exists x\ (P(x) \wedge Q(x))$
- Answer 1:Only 1
- Answer 2:Only 2
- Answer 3:Both 1. and 2.